

ROLANDO S. GARCIA, PH.D.

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EDUCATION

University of California, Berkeley Ph.D. in Computer Science advised by Dr. Joseph M. Hellerstein Dissertation: <i>The Management of Context in the Machine Learning Lifecycle</i>	Berkeley, CA 2017 – 2024
Arizona State University B.S. in Computer Science, <i>Summa Cum Laude</i>	Tempe, AZ 2014 – 2017
Northern Arizona University Philosophy and Computer Information Systems	Flagstaff, AZ 2010 – 2013

AWARDS AND FELLOWSHIPS

ASU Presidential Postdoctoral Fellowship, two years of funding.	2024
Dell Technologies Inc. Research Award, one year of funding.	2023
NSF Graduate Research Fellowship, three years of funding.	2017
UC Berkeley Chancellor's Fellowship, two years of funding.	2017
UC Berkeley's EECS Excellence Award.	2017
CRA Outstanding Undergraduate Researcher Award, honorable mention.	2017
Fulton Undergraduate Research Initiative, research award.	2016
IEEE VAST Grand Challenge Award, outstanding comprehensive submission.	2015

RESEARCH

School of Computing & AI Presidential Postdoctoral Fellow, advised by <u>Dr. Ross Maciejewski</u> - Submitted and under review: - NSF TTP-T: <i>Scalable Human-in-the-Loop Review: Expert-AI Collaboration in High-Volume Document Processing</i>, w/ R. Maciejewski, & T. Winarski¹ - Co-PI at 40% effort, \$759k over 3 years. - NSF Future CoRE: <i>III: Explainable GNN Approaches to Ad-Hoc Queries on Heterogeneous Networks</i>, w/ Yixuan He² et al. - Co-PI at 20% effort, \$1M over 3 years.	July 2024 – Present Arizona State University
RISE Lab rise.cs.berkeley.edu Graduate Student Researcher & Fellow, advised by <u>Dr. Joe Hellerstein</u> - FlorDB: a hindsight logging database for the machine learning lifecycle. ... How Engineers Operationalize Machine Learning, w/ Shreya Shankar et al. - Internships: Linea Labs (<i>startup</i>).	June 2017 – June 2024 University of California, Berkeley github.com/ucbrise/flor
Amazon A9 (Search) Research Intern, Search Infrastructure Team - Studied end-to-end ML pipelines orchestrated via AWS SageMaker, spanning ingestion, transformation, tokenization, training, and evaluation across multi-team workflows. - Observed and contributed to DAG-based pipeline design for large-scale search model training and A/B testing.	May 2018 – December 2018 Palo Alto, CA

¹ Sandra Day O'Connor College of Law

² School of Mathematical and Natural Sciences

PUBLICATIONS

- Citation Analysis and Deterministic Evaluation at Scale** *Submitted, TIIS 2026*
Rolando Garcia, Akshit Jain, Shiyi Liu, Rostyslav Hnatyshyn, Fan Lei, Ross Maciejewski.
ACM Transactions on Interactive Intelligent Systems, under review.
- IDSPLACE: A Novel Document Generator for Reliable Evaluation of Digital Identity Verification Systems** *Submitted, ICDM 2026*
Lulu Xie, Yancheng Wang, Kanchan Chowdhury, Rolando Garcia, Yingzhen Yang, Jia Zou.
IEEE International Conference on Data Mining, under review.
- Cost-Benefit Modeling of Interactive Visual Machine Learning Systems** *IEEE VIS 2026*
Yixuan Wang, Rolando Garcia, Alex Endert, Ross Maciejewski.
IEEE Transactions on Visualization and Computer Graphics, 2026.
- Flow with FlorDB: Incremental Context Maintenance for the ML Lifecycle** *CIDR 2025*
Rolando Garcia, Pragya Kallanagoudar, Chithra Anand, Sarah E. Chasins, Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein.
Conference on Innovative Data Systems Research, 2025.
- “We Have No Idea How Models will Behave in Production until Production”: How Engineers Operationalize Machine Learning** *CSCW 2024*
Shreya Shankar*, Rolando Garcia*, Joseph M. Hellerstein, Aditya G. Parameswaran.
ACM Conf’ on Computer Supported Cooperative Work and Social Computing 2024.
*Co-first authors. **Honorable Mention: MLOps Community Best Paper Award**
- Quilt: Custom UIs for Linking Unstructured Documents to Structured Dataasets** *UIST Demo 2024*
Pragya Kallanagoudar, Chithra Anand, Rolando Garcia, Rebecca M. M. Hicke, Aditya G. Parameswaran, Eunice Jun, Sarah E. Chasins. *UIST Adjunct ’24: Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 2024.
- Hindsight Logging for Model Training** *VLDB 2021*
Rolando Garcia, Eric Liu, Vikram Sreekanti, Bobby Yan, Anusha Dandamudi, Joseph E. Gonzalez, Joseph M. Hellerstein, Koushik Sen. *Proceedings of the VLDB Endowment*, 14(4): 682-693, 2021.
- Context: The Missing Piece in the Machine Learning Lifecycle** *KDD CMI 2018*
Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Joseph E. Gonzalez, Joseph M. Hellerstein.
Workshop on Common Model Infrastructure at KDD, 2018.
- A Visual Analytics Framework for Exploring Theme Park Dynamics** *ACM TIIS 2018*
Michael Steptoe, Robert Krüger, Rolando Garcia, Xing Liang, Ross Maciejewski.
ACM Transactions on Interactive Intelligent Systems, 2018.
- The State-of-the-Art in Predictive Visual Analytics** *CGF 2017*
Yafeng Lu, Rolando Garcia, Brett Hansen, Michael Gleicher, Ross Maciejewski.
Computer Graphics Forum, 2017.

TALKS

- Incremental Context Maintenance for the ML Lifecycle** *CIDR 2025*
Rolando Garcia, Pragma Kallanagoudar, Chithra Anand, Sarah E. Chasins,
Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein.
Conference on Innovative Data Systems Research. January 2025. Amsterdam
- How Engineers Operationalize Machine Learning** *CSCW 2024*
Shreya Shankar, *Rolando Garcia*, Joseph M. Hellerstein, Aditya G. Parameswaran.
ACM Conf' on Computer Supported Cooperative Work and Social Computing.
November 2024. Costa Rica
- Flow with FlorDB: Agile MLOps without Compromises** *Mines Seminar, 2024*
Rolando Garcia. Colorado School of Mines. October 2024. Golden, CO
- Operationalizing Machine Learning: An Interview Study** *Flyte, 2022*
Rolando Garcia, Shreya Shankar. December 2022.
- Operationalizing Machine Learning: An Interview Study** *Inaugural Address, 2022*
Shreya Shankar, *Rolando Garcia*. Berkeley, CA, October 2022. EPIC, UC Berkeley
- Operationalizing Machine Learning: An Interview Study** *Canva, 2022*
Shreya Shankar, *Rolando Garcia*. October 2022.
- Operationalizing Machine Learning: An Interview Study** *MLOps World, 2022*
Shreya Shankar, *Rolando Garcia*. San Francisco Bay Area, October 2022. San Francisco
- FLOR: Managing the Infinite Virtual Metadata of Model Training** *Cornell Seminar, 2022*
Rolando Garcia. Cornell University, February 2022.
- Hindsight Logging for Model Training** *VLDB 2021*
Rolando Garcia, Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan,
Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. *International Conference*
on Very Large Databases. August 2021. Copenhagen
- Choose What to Log, After you Execute** *RISE Retreat 2019*
Rolando Garcia, Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan,
Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. Retreat talk to lab sponsors.
Monterey, CA. January 2019.
- ML in Context: Preserve, Share, and Analyze Experiments with Flor** *RISE Camp 2018*
Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw,
Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein.
Talk and Tutorial for lab sponsors. Berkeley, CA. October 2018.
- Context: The Missing Piece in the Machine Learning Lifecycle** *KDD 2018*
Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw,
Joseph Gonzalez, Joseph Hellerstein. *The 24th ACM SIGKDD Intl' Conf*. August 2018. London
- Data Management Challenges in the Machine Learning Lifecycle** *Amazon 2018*
Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar,
Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein.
Invited talk at Amazon Search. Palo Alto, CA. April 2018. Palo Alto

Design, Run, and Reproduce Experiments with Jarvis*RISE Retreat* 2018

Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar,
Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein.

Retreat talk to lab sponsors. Monterey, CA. January 2018.

Jarvis: Managing the Machine Learning Lifecycle*Google* 2017

Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar,
Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein.

Mountain View

Invited talk at Google. Mountain View, CA. September 2017.

The State-of-the-Art in Predictive Visual Analytics*EuroVis* 2017

Yafeng Lu, *Rolando Garcia*, Brett Hansen, Michael Gleicher, Ross Maciejewski.

Barcelona

19th *EG/VGTC Conference on Visualization*, Barcelona, Spain. June 2017.

Automatic Design and Optimization of Deep Learning Models*LANS* 2016

Rolando Garcia, Randal Olson, Jason Moore.

The Leadership Alliance National Symposium, Stamford, CT. July 2016.

VAST 2015 Grand Challenge – Team VADER/VIS*VAST* 2015

Michael Steptoe, Robert Krüger, Yifan Zhang, Xing Liang, *Rolando Garcia*,

Chicago

Sagarika Kadambi, Wei Luo, Thomas Ertl, Ross Maciejewski.

IEEE Visual Analytics Science and Technology Conference, Chicago, IL. October 2015.

TEACHING**Introduction to Database Systems (COMPSCI 186/286)**

Spring 2018

Graduate Student Instructor, 450 students

*University of California, Berkeley***Introduction to Programming Languages (CSE 240)**

Fall 2015

Undergraduate Student Instructor, 267 students

*Arizona State University, Tempe***Introduction to Ethics (PHI 105)**

Fall 2013

Undergraduate Student Instructor, 50 students

Northern Arizona University, Flagstaff

SERVICE**Diversifying Access to Research in Engineering (DARE)**

June 2020 – December 2022

Organizer | dare.berkeley.edu

*University of California, Berkeley***EECS Graduate Admissions**

Fall 2021

Graduate Student Representative | eecs.berkeley.edu

*University of California, Berkeley***Data Systems and Foundations Seminar**

Fall 2020

Coordinator | dsf.berkeley.edu

University of California, Berkeley